Polar Form, De Moivre's Theorem & Roots of Unity

ANSWER KEY

Converting to polar (modulus-argument) form

1 Write $z = 1 + \sqrt{3}i$ in polar form $r(\cos\theta + i\sin\theta)$, giving the argument in radians.
$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$. $\theta = \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$ (first quadrant). Polar form: $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$.

2 Convert $z = 4\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$ to Cartesian form $a + bi$.
$a = 4\cos\left(\frac{5\pi}{6}\right) = 4\left(-\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$. $b = 4\sin\left(\frac{5\pi}{6}\right) = 4\left(\frac{1}{2}\right) = 2$. So $z = -2\sqrt{3} + 2i$.

Multiplying and dividing in polar form

3 Given $z_1 = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ and $z_2 = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, find z_1z_2 in polar form.
Multiply moduli, add arguments: $z_1z_2 = 3(2)\left(\cos\left(\frac{\pi}{4}+\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{4}+\frac{\pi}{6}\right)\right) = 6\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$.

4 Using the same z_1 and z_2, find $\frac{z_1}{z_2}$ in polar form.
Divide moduli, subtract arguments: $\frac{z_1}{z_2} = \frac{3}{2}\left(\cos\left(\frac{\pi}{4}-\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{4}-\frac{\pi}{6}\right)\right) = \frac{3}{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$.

De Moivre's theorem

5 State De Moivre's theorem, and use it to find $\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^4$.
De Moivre's theorem: $[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$. Applying it: $2^4\left(\cos\frac{4\pi}{6} + i\sin\frac{4\pi}{6}\right) = 16\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 16\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -8 + 8\sqrt{3}i$.

6 Use De Moivre's theorem to find $(1+i)^6$, giving your answer in Cartesian form.
Convert to polar: $|1+i| = \sqrt{2}$, $\arg(1+i) = \frac{\pi}{4}$. So $(1+i)^6 = (\sqrt{2})^6\left(\cos\frac{6\pi}{4} + i\sin\frac{6\pi}{4}\right) = 8\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = 8(0 - i) = -8i$.

Finding roots of complex numbers

7 Find the three cube roots of $z = 8\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.
The cube roots have modulus $8^{1/3} = 2$ and arguments $\frac{\pi/2 + 2\pi k}{3}$ for $k = 0, 1, 2$: $\frac{\pi}{6}$, $\frac{\pi}{6} + \frac{2\pi}{3} = \frac{5\pi}{6}$, and $\frac{\pi}{6} + \frac{4\pi}{3} = \frac{3\pi}{2}$. So the roots are $2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$, $2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$, and $2\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$.

The nth roots of unity

8 Find all four fourth roots of unity (solutions to $z^4 = 1$), and explain their geometric arrangement on the Argand diagram.

Writing $1 = \cos 0 + i\sin 0$, the roots have modulus 1 and arguments $\frac{2\pi k}{4}$ for $k = 0, 1, 2, 3$: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, giving $z = 1, i, -1, -i$. Geometrically, these four points lie evenly spaced around the unit circle, forming the vertices of a square.

Using De Moivre's theorem to derive trig identities

9 Use De Moivre's theorem with $n = 2$ to express $\cos 2\theta$ in terms of $\cos\theta$ and $\sin\theta$, by expanding $(\cos\theta + i\sin\theta)^2$ and comparing real parts.

$(\cos\theta + i\sin\theta)^2 = \cos^2\theta + 2i\cos\theta\sin\theta + i^2\sin^2\theta = (\cos^2\theta - \sin^2\theta) + i(2\sin\theta\cos\theta)$. By De Moivre's theorem, this also equals $\cos 2\theta + i\sin 2\theta$. Comparing real parts: $\cos 2\theta = \cos^2\theta - \sin^2\theta$.

10 Using the same expansion from the previous problem, compare imaginary parts to find an identity for $\sin 2\theta$.

Comparing imaginary parts of $(\cos^2\theta - \sin^2\theta) + i(2\sin\theta\cos\theta) = \cos 2\theta + i\sin 2\theta$: $\sin 2\theta = 2\sin\theta\cos\theta$.

Roots of a general complex number

11 Find the two square roots of $z = -4$, giving your answers in Cartesian form.

Write $-4 = 4(\cos\pi + i\sin\pi)$. Square roots have modulus $4^{1/2} = 2$ and arguments $\frac{\pi + 2\pi k}{2}$ for $k = 0, 1$: $\frac{\pi}{2}$ and $\frac{3\pi}{2}$. Roots: $2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 2i$, and $2\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) = -2i$.

12 Solve the equation $z^3 = -8i$, giving all three roots in polar form with arguments in $[0, 2\pi)$.

Write $-8i = 8\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Cube roots have modulus $8^{1/3} = 2$ and arguments $\frac{3\pi/2 + 2\pi k}{3}$ for $k = 0, 1, 2$: $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$. Roots: $2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$, $2\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$, $2\left(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}\right)$.

Applying De Moivre's theorem to negative and fractional powers

13 Find z^{-3} for $z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$, using the extended form of De Moivre's theorem for negative integer powers.

De Moivre's theorem extends to negative integers: $z^{-3} = 2^{-3}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right) = \frac{1}{8}\left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) = -\frac{\sqrt{2}}{16} - \frac{\sqrt{2}}{16}i$.

14 Sum the four fourth roots of unity found earlier in this worksheet ($1, i, -1, -i$), and explain the general result this illustrates about the sum of all nth roots of unity for $n \geq 2$.

$1 + i + (-1) + (-i) = 0$. In general, the sum of all n distinct nth roots of unity is always 0 for $n \geq 2$, since the roots are evenly and symmetrically spaced around the unit circle, causing their vector sum to cancel completely.

Solving equations that reduce to roots of unity

15 Solve $(z+1)^4 = 1$ for z, by first finding the fourth roots of unity and then solving for z.

The fourth roots of unity are $w = 1, i, -1, -i$ (found earlier). Setting $z+1 = w$: $z = w-1$, giving $z = 0,\ i-1,\ -2,\ -i-1$.

16 Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and standard polar division for the quotient.

$1+i$ has modulus $\sqrt{2}$ and argument $\frac{\pi}{4}$, so $(1+i)^3$ has modulus $(\sqrt{2})^3 = 2\sqrt{2}$ and argument $\frac{3\pi}{4}$. $1-i$ has modulus $\sqrt{2}$ and argument $-\frac{\pi}{4}$. Dividing: modulus $= \frac{2\sqrt{2}}{\sqrt{2}} = 2$; argument $= \frac{3\pi}{4} - \left(-\frac{\pi}{4}\right) = \pi$. So z has modulus 2 and argument π.

Using De Moivre's theorem for a triple-angle identity

17 Use De Moivre's theorem with $n = 3$ to express $\cos 3\theta$ in terms of $\cos\theta$, by expanding $(\cos\theta + i\sin\theta)^3$ and comparing real parts.

$(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. The real part is $\cos^3\theta - 3\cos\theta\sin^2\theta$. By De Moivre's theorem, this equals $\cos 3\theta$. Substituting $\sin^2\theta = 1-\cos^2\theta$: $\cos 3\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

18 Find the six sixth roots of unity, and state which of these roots are also fourth roots of unity.

The sixth roots of unity have modulus 1 and arguments $\frac{2\pi k}{6} = \frac{\pi k}{3}$ for $k = 0, 1, 2, 3, 4, 5$: $0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$, giving $z = 1,\ \frac{1}{2} + \frac{\sqrt{3}}{2}i,\ -\frac{1}{2} + \frac{\sqrt{3}}{2}i,\ -1,\ -\frac{1}{2} - \frac{\sqrt{3}}{2}i,\ \frac{1}{2} - \frac{\sqrt{3}}{2}i$. The fourth roots of unity are $1, i, -1, -i$. Comparing the two lists, only $z = 1$ and $z = -1$ appear in both — these are the roots common to both sets, since they correspond to arguments that are multiples of both $\frac{\pi}{2}$ and $\frac{\pi}{3}$ (i.e. multiples of π).

Applying De Moivre's theorem to solve an equation with a shifted variable

19 Solve $z^5 = 32$ for z, giving all five roots in polar form with arguments in $[0, 2\pi)$.

Write $32 = 32(\cos 0 + i\sin 0)$. The fifth roots have modulus $32^{1/5} = 2$ and arguments $\frac{2\pi k}{5}$ for $k = 0, 1, 2, 3, 4$: $0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$. Roots: $2\left(\cos\frac{2\pi k}{5} + i\sin\frac{2\pi k}{5}\right)$ for $k = 0, 1, 2, 3, 4$, i.e. $2,\ 2\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right),\ 2\left(\cos\frac{4\pi}{5} + i\sin\frac{4\pi}{5}\right),\ 2\left(\cos\frac{6\pi}{5} + i\sin\frac{6\pi}{5}\right),\ 2\left(\cos\frac{8\pi}{5} + i\sin\frac{8\pi}{5}\right)$.

Combining polar arithmetic with De Moivre's theorem

20 Given $z_1 = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$, find z_1^8 in its simplest Cartesian form, and explain why the result is a positive real number.

By De Moivre's theorem: $z_1^8 = (\sqrt{2})^8\left(\cos\frac{8\pi}{4} + i\sin\frac{8\pi}{4}\right) = 16(\cos 2\pi + i\sin 2\pi) = 16(1 + 0i) = 16$. The result is a positive real number because the argument $\frac{8\pi}{4} = 2\pi$ is a full rotation, landing back on the positive real axis where $\sin = 0$ and $\cos = 1$.